

CBSE 2027 Maths — Predicted Paper (Set 8 of 10)

Based on the highly weighted CBSE syllabus chapters and a rigorous analysis of Previous Year Question (PYQ) trends, here is Set 8 of the high-probability predicted Class 12 Maths board paper. This predicted paper focuses heavily on Calculus, Vectors, 3D Geometry, and Competency-Based Case Studies.

SECTION A: Objective Type Questions (1 Mark Each x 20 = 20 Marks)

Q1. Let R be a relation on the set of natural numbers $\mathbb{N}$ defined by $x + 2y = 8$. What is the domain of the relation R ? * **Chapter:** Relations and Functions * **Probability Tag: Medium**

Q2. Find the principal value of $\sin^{-1}(\sin \frac{3\pi}{5})$. * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: Very High**

Q3. If A is a square matrix of order 3×3 such that $|A| = 5$, find the exact value of $|A^2|$. * **Chapter:** Determinants * **Probability Tag: High**

Q4. Given that A is a non-singular matrix of order 3, and $A^2 = 2A$, what is the value of the determinant $|A|$? * **Chapter:** Matrices and Determinants * **Probability Tag: High**

Q5. What is the derivative of $\log_e(\log_e x)$ with respect to x , where $x > 1$? * **Chapter:** Continuity and Differentiability * **Probability Tag: Medium**

Q6. The radius of a spherical soap bubble is increasing at the rate of 0.2 cm/s . What is the rate of increase of its surface area when the radius is 5 cm ? * **Chapter:** Application of Derivatives * **Probability Tag: Very High**

Q7. Evaluate the definite integral: $\int_0^{\pi/2} \frac{dx}{1+\tan x}$. * **Chapter:** Integrals * **Probability Tag: Very High**

Q8. Find the area of the region bounded by the curve $y = 2x^2$, the x-axis, and the vertical lines $x = 1$ and $x = 3$. * **Chapter:** Application of Integrals * **Probability Tag: Medium**

Q9. Determine the order and degree (if defined) of the differential equation: $\left(\frac{d^3y}{dx^3}\right)^2 + \frac{dy}{dx} = e^x$. * **Chapter:** Differential Equations * **Probability Tag: Very High**

Q10. Find a vector of magnitude 5 units parallel to the resultant of the vectors $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$. * **Chapter:** Vector Algebra * **Probability Tag: High**

Q11. If $|\vec{a}| = 2$, $|\vec{b}| = 3$, and $\vec{a} \cdot \vec{b} = 4$, find the magnitude of $|\vec{a} - \vec{b}|$. * **Chapter:** Vector Algebra * **Probability Tag: Very High**

Q12. Write the direction cosines of a straight line which is equally inclined to the negative directions of the x, y, and z axes. * **Chapter:** Three-Dimensional Geometry * **Probability Tag: High**

Q13. Find the vector equation of a plane which is at a perpendicular distance of 6 units from the origin and has its normal vector as $2\hat{i} - \hat{j} + 2\hat{k}$. * **Chapter:** Three-Dimensional Geometry

* **Probability Tag: High**

Q14. In a given Linear Programming Problem, the feasible region has corner points $(0, 0)$, $(5, 0)$, $(3, 4)$, and $(0, 6)$. Find the minimum value of the objective function $Z = 4x + 6y$. *

Chapter: Linear Programming * **Probability Tag: Very High**

Q15. If $P(A) = 0.4$, $P(B) = 0.5$, and the conditional probability $P(B|A) = 0.6$, evaluate $P(A \cup B)$. * **Chapter:** Probability * **Probability Tag: High**

Q16. Evaluate the indefinite integral: $\int \frac{\cos 2x}{(\sin x + \cos x)^2} dx$. * **Chapter:** Integrals * **Probability Tag: Very High**

Q17. Find the integrating factor of the linear differential equation: $x \frac{dy}{dx} - 2y = x^2$. * **Chapter:** Differential Equations * **Probability Tag: High**

Q18. A random variable X takes the values $0, 1, 2, 3$ with respective probabilities $k, 3k, 3k, k$. Find the exact value of k . * **Chapter:** Probability * **Probability Tag: Medium**

Q19. (Assertion-Reason): Assertion (A): The function $f(x) = |x|$ is continuous everywhere but not differentiable at $x = 0$. **Reason (R):** Every continuous function is automatically a differentiable function. * **Chapter:** Continuity and Differentiability * **Probability Tag: High**

Q20. (Assertion-Reason): Assertion (A): The vectors $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ and $\vec{b} = -4\hat{i} + 6\hat{j} - 8\hat{k}$ are parallel to each other. **Reason (R):** Two non-zero vectors $\vec{a}$ and $\vec{b}$ are parallel if $\vec{a} = \lambda\vec{b}$ for some non-zero scalar λ . * **Chapter:** Vector Algebra * **Probability Tag: Very High**

SECTION B: Very Short Answer Type Questions (2 Marks Each x 5 = 10 Marks)

Q21. Prove that $2 \sin^{-1} \left(\frac{3}{5} \right) = \tan^{-1} \left(\frac{24}{7} \right)$. * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: Very High**

Q22. If $y = A \sin x + B \cos x$, prove that $\frac{d^2y}{dx^2} + y = 0$. * **Chapter:** Continuity and Differentiability * **Probability Tag: High**

Q23. Let $\hat{a}$ and $\hat{b}$ be two unit vectors and θ be the angle between them. Show that $\sin\left(\frac{\theta}{2}\right) = \frac{1}{2}|\hat{a} - \hat{b}|$. * **Chapter:** Vector Algebra * **Probability Tag: Very High**

Q24. Solve the differential equation: $\frac{dy}{dx} = 2^{x+y}$. * **Chapter:** Differential Equations * **Probability Tag: High**

Q25. Given that the events E and F are such that $P(E) = \frac{1}{2}$, $P(E \cup F) = \frac{3}{5}$, and $P(F) = x$. Find the value of x if E and F are independent events. * **Chapter:** Probability * **Probability Tag: High**

SECTION C: Short Answer Type Questions (3 Marks Each x 6 = 18 Marks)

Q26. Let $A = \{1, 2, 3, \dots, 9\}$. Let R be the relation on $A \times A$ defined by $(a, b)R(c, d)$ if and only if $a + d = b + c$. Prove that R is an equivalence relation. * **Chapter:** Relations and Functions * **Probability Tag: Very High**

Q27. If $A = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$, find $f(A)$ where the polynomial is $f(x) = x^2 - 4x + 1$. Show that $f(A) = O$ (the null matrix). * **Chapter:** Matrices * **Probability Tag: High**

Q28. Differentiate the function $\sin^2 x$ with respect to $e^{\cos x}$. * **Chapter:** Continuity and Differentiability * **Probability Tag: Very High**

Q29. Evaluate the definite integral using standard properties of limits: $\int_0^{\pi/2} \log(\sin x) dx$. * **Chapter:** Integrals * **Probability Tag: Very High**

Q30. Solve the linear differential equation: $(1 + y^2)dx = (\tan^{-1} y - x)dy$. * **Chapter:** Differential Equations * **Probability Tag: Very High**

Q31. An urn contains 5 red balls and 2 black balls. Two balls are drawn at random without replacement. Let X represent the number of black balls drawn. Find the mean (expected value) of X . * **Chapter:** Probability * **Probability Tag: High**

SECTION D: Long Answer Type Questions (5 Marks Each x 4 = 20 Marks)

Q32. Using the method of integration, find the area bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. * **Chapter:** Application of Integrals * **Probability Tag: High**

Q33. Find the equation of the plane passing through the points $(2, 1, -1)$ and $(-1, 3, 4)$ and which is perpendicular to the plane $x - 2y + 4z = 10$. * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Very High**

Q34. Use the matrix product $\begin{pmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{pmatrix} \begin{pmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{pmatrix}$ to solve the following

system of linear equations: $x - y + 2z = 1$ $2y - 3z = 1$ $3x - 2y + 4z = 2$ * **Chapter:** Matrices and Determinants * **Probability Tag: Very High**

Q35. A wire of length **28 m** is to be cut into two pieces. One of the pieces is to be bent into the shape of a square and the other into the shape of a circle. What should be the exact length of the two pieces so that the combined area of the square and the circle is minimum? * **Chapter:** Application of Derivatives * **Probability Tag: Very High**

SECTION E: Case-Study Based Questions (4 Marks Each x 3 = 12 Marks)

Q36. Case Study 1: Matrices (School Library Revenue) A prominent school library has an extensive collection of textbooks for senior secondary classes. The librarian takes an inventory and finds that there are 20 dozen chemistry books, 15 dozen physics books, and 10 dozen economics books in the reference section. The school decides to sell the surplus stock

to students at discounted selling prices of ₹80, ₹60, and ₹40 each respectively. (i) Represent the total number of individual books of each subject in a row matrix. (1 Mark) (ii) Represent the selling prices of the books in a column matrix. (1 Mark) (iii) Using the principles of matrix multiplication, find the total amount the library will receive from selling all of these surplus books. (2 Marks) * **Chapter:** Matrices * **Probability Tag: Very High**

Q37. Case Study 2: Linear Programming (Manufacturing Logistics) A manufacturing company creates two models, Model A and Model B, of a specialized electronic product. Each piece of Model A requires 9 labour hours for fabricating and 1 labour hour for finishing. Each piece of Model B requires 12 labour hours for fabricating and 3 labour hours for finishing. For the upcoming week, the maximum labour hours available for fabricating and finishing are 180 hours and 30 hours respectively. The company makes a profit of ₹8000 on each piece of Model A and ₹12000 on each piece of Model B. (i) Formulate the mathematical constraints for the labour hours in fabricating and finishing. (1 Mark) (ii) Write the objective function Z to maximize the total profit. (1 Mark) (iii) Determine the exact number of pieces of Model A and Model B that should be manufactured to secure the maximum profit, and calculate that profit. (2 Marks) * **Chapter:** Linear Programming * **Probability Tag: Very High**

Q38. Case Study 3: Probability (COVID-19 Testing & Bayes' Theorem) During a localized outbreak, the reliability of a COVID-19 rapid test is specified as follows: Of the people actually having COVID-19, the test is 90% accurate in detecting the disease but 10% go undetected. Of the people completely free of COVID-19, 99% of the tests are correctly judged negative but 1% are falsely diagnosed as showing COVID positive (a false positive). From a large local population of which only 0.1% genuinely have COVID-19, one person is selected at random and given the rapid test. (i) Let E_1 be the event of having the disease and A be the event of testing positive. What is the conditional probability $P(A|E_1)$? (1 Mark) (ii) Use the Law of Total Probability to find the total probability that the randomly selected person tests positive. (1 Mark) (iii) If the pathologist reports the person as COVID positive, use Bayes' theorem to find the exact probability that the person *actually* has the disease. (2 Marks) * **Chapter:** Probability * **Probability Tag: Very High**

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.